CoRec Muscle Finder Project Design

CS193 Final Project Planning

This plan describes a desktop app for finding CoRec machines and exercises by muscle group. The first version focuses on muscle selection, machine summaries and locations, and exercise details.

1 1 Problem Statement

When I work out at the CoRec, the machines I want to use are often occupied, and I do not always know which other machines can train the same muscle. Finding an alternative interrupts my workout and can leave me waiting for equipment. I want an easy way to find other suitable machines and their locations so I can keep my workout moving.

1 2 Project Idea

I will build a Java desktop application that lets users select a muscle group and see matching CoRec machines with short summaries and locations. Clicking a machine will show exercises for the selected muscle.

2 2 Tech Stack

Language: Java for application logic.

Interface: JavaFX for muscle selection, machine results, and exercise details.

Libraries and build tools: Gson for parsing JSON and Maven for managing dependencies and builds.

Data storage: A local JSON file containing machine names, summaries, CoRec locations, target muscles, and exercises. No database is needed.

Data sources: I will manually record and verify machine locations at the CoRec and curate a small exercise collection with source references. No external API or live occupancy data is required.

Development tools: IntelliJ IDEA for coding and Git/GitHub for version control and documentation.

2 1 Eisenhower Matrix

The Do quadrant defines the MVP. Other features are deferred, handled by existing libraries, or removed from scope.

	Urgent	Not Urgent
Important	**Do - Your MVP** • Select a target muscle group. • Show matching machines with summaries and CoRec locations. • Click a machine to see exercises for the selected muscle.	**Schedule - V2 and Later** • Add a clickable body diagram for muscle selection. • Add exercise pictures and more detailed instructions. • Add a visual gym map of machine locations. • Explore a phone-friendly version.
Not Important	**Delegate - Use a Library** • Use JavaFX controls and layouts for the interface. • Use Gson to parse the JSON data. • Use Java standard file utilities to read local data.	**Delete - Cut It** • Automatic live machine occupancy detection. • Favorites, workout logging, and equipment availability filters. • Social profiles, followers, and workout sharing. • AI-generated personalized workout plans. • Payments and subscription features.

2 3 Chunked Build Plan

Each milestone will end with one meaningful Git commit after its behavior has been checked.

1. Configure the Maven project with JavaFX and Gson, and verify that a basic window opens.
2. Create a local JSON collection of machines, summaries, locations, target muscles, and exercises.
3. Create Java data classes and load the JSON using Gson; verify that the expected records load.
4. Add a muscle selection menu and matching logic; check that each selection returns the correct machines.
5. Display matching machine summaries and locations, including a helpful message when there are no matches.
6. Make machines clickable and show exercises matching both the selected machine and muscle.
7. Polish the interface, check the complete workflow with different muscle selections, and fix issues found.
8. Finish the README with setup and run instructions, screenshots, and known limitations.

3 1 Public GitHub Repository

https://github.com/Rafael-Lindemberg/corec-muscle-finder

The repository will contain the README, the Java source and resource folders, this design document, the matrix PDF, and both the editable system diagram and its exported image.

2 4 System Diagram

The app reads a local JSON file to find machines for the selected muscle, display their summaries and locations, and show exercises after a machine is selected.

